

SWAYAM MEHTA

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EDUCATION

M.S. Computer Science	2024 - 2026
Arizona State University	4.0/4.0 GPA
B.Tech. Computer Science and Engineering	2020 - 2024
Vellore Institute of Technology	8.65/10.0 GPA

SKILLS

Programming	Python, Java, Kotlin, JavaScript, TypeScript, SQL, C++
Frameworks/Libraries	FastAPI, Flask, React, Next.js, D3.js, REST, PyTorch
Tools/Technologies	Linux, Git, PostgreSQL, Redis, Docker, Kubernetes, AWS, Prometheus, Grafana

EXPERIENCE

JobTalk AI: *Machine Learning Engineering Intern* 03/2026 - 05/2026

- Built an async multi-format resume ingestion service (FastAPI, Celery, MinIO/S3) with a 3-tier parallel extraction pipeline, AES-128 PII-at-rest encryption and row-level tenant isolation for a multi-client SaaS deployment.
- Diagnosed and resolved a Celery/asyncpg connection-pool exhaustion bug failing resumes under load, lifting end-to-end success to **99%+**; deployed Prometheus + Grafana monitoring across 19 panels tracking throughput, p50/p95/p99 latency, and queue depth across the pipeline.

Vellore Institute of Technology: *Research Assistant* 01/2024 - 05/2024

- Improved a stock price forecasting pipeline (Bi-LSTM, GRU, GAN); delivered forecasts and VADER sentiment scores via a Flask REST API with a Gemini Vision chart-analysis chatbot, reducing RMSE by **40%** over prior baselines.

Hewlett Packard Enterprise: *CTY Intern - High Performance Computing* 01/2023 - 06/2023

- Engineered a Kubernetes runtime security system using Falco with eBPF probes to detect container escapes, privilege escalation, and filesystem tampering; streamed alerts via Falco Exporter over gRPC into Prometheus/Grafana with Slack and Discord notifications.
- Increased security coverage by **~30%** and cut incident detection latency by **~40%** in attack simulations by replacing manual log review with automated rule-based behavioral alerts.

PROJECTS

ClickLess AI [Python, LangGraph, Mistral, Neo4j, Playwright, Browser Use, Streamlit] Spring 2026

- Architected a 4-stage multi-agent pipeline converting a natural language grocery request into a ranked cart submission, orchestrating Mistral 7B NLU, Instacart API retrieval, knowledge-graph ranking, and Playwright checkout; delivering a **4.2s** time-to-first-option and eliminating **11.8** manual interactions per session.
- Engineered a credential system that stores retailer login data in an encrypted vault and injects short-lived session tokens at checkout, preventing credential replay, verified across **97%** of live checkout sessions.

Quizzi [Next.js, TypeScript, PostgreSQL, Supabase, Redis, D3.js] Summer 2025

- Designed an indexed PostgreSQL schema with row-level security; eliminated **47 RLS performance warnings** by replacing volatile row-evaluations with a subquery constant, reducing policy evaluation from $O(N)$ to $O(1)$ and achieving **10–100x** query speedup on large datasets.
- Reduced daily database egress by **76%** by decomposing heavy LATERAL JOINS into lightweight primary-key queries with client-side assembly, layering TTL caching and 50ms debounce batching to collapse N+1 profile fetches.

Industrial Ops Monitor [Python, FastAPI, React, TypeScript, Gemini, SSE] Spring 2026

- Built a full-stack equipment intelligence platform over **3M real sensor readings** across 96 assets with a **3-pass anomaly engine** (median deviation, threshold breach, linear-regression trend projection), SSE-streamed Gemini briefings, and an automated work-order lifecycle, validated on a real bearing failure **35 days** before it occurred.

LEADERSHIP

IEEE Computer Society - VIT: *Secretary* 01/2023 - 12/2023

- Spearheaded a cross-functional team to launch Chota Dhobi, a laundry-tracking app with **24K+ active users**, driving **15% month-over-month growth** through targeted onboarding campaigns.